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Studierichting: Informatica
Oefeningenreeks 6E nr. 2.3

$T_n\left(f,0\right)(x)$ voor $f(x)=\cos x$ en bewijs $\lim\limits_{n\to+\infty} R_n(x)=0$

1. $T_n\left(\cos x,0\right)(x)$

   $f(x)=\cos x$

   $f'(x)=-\sin x$

   $f''(x)=-\cos x$

   $f'''(x)=\sin x$

   $f^{iv}(x)=\cos x$

   $f(0)=1$

   $f'(0)=0$

   $f''(0)=-1$

   $f'''(0)=0$

   $f^{iv}(0)=1$

   $\Rightarrow f^{(2k+1)}(0)=0$ en $f^{(2k)}(0)=(-1)^k$

   $$\Rightarrow T_n\left(\cos x,0\right)(x)=\sum_{k=0}^{\infty}\frac{f^{(k)}(0)}{k!}x^k$$

   Alleen even termen blijven over:

   $$=\sum_{k=0}^{\infty}\frac{f^{(2k)}(0)}{(2k)!}x^{2k}=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!}x^{2k}$$

2. Bewijs $\lim\limits_{n\to+\infty} R_n(x)=0$

   $$R_n(x)=\frac{f^{(2n+2)}(\xi)}{(2n+2)!}x^{2n+2}\quad \xi\in[0,x]$$

$\left|f^{(m)}(x)\right| \leq 1$

$|R_n(x)| \leq \dfrac{1}{(2n+2)!}|x|^{2n+2}$

Gebruik D'Alembert: Beschouw $\displaystyle\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

$\displaystyle\lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to+\infty} \frac{x^{2n+2}}{(2n+2)!} \cdot \frac{(2n)!}{x^{2n}}$

$\displaystyle = \lim_{n\to+\infty} \frac{x^2}{(2n+2)(2n+1)} = \frac{x^2}{\infty} = 0 \Rightarrow$ De reeks is convergent.

Bijgevolg moet $\displaystyle\lim_{n\to+\infty} R_n(x) = \lim_{n\to+\infty} \frac{x^{2n+2}}{(2n+2)!} = 0$

References